

An approach to portfolio optimization with time series forecasting algorithms and machine learning techniques

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ABSTRACT

The challenge of identifying suitable stocks for portfolio inclusion, particularly in the context of complex stock forecasting dynamics characterized by nonlinear time series and various influencing factors, is addressed in this study. To tackle this challenge, an approach combining the auto-regressive integrated moving average (ARIMA) and least-square support vector machine (LS-SVM) models is proposed for stock selection. Furthermore, the mean-variance portfolio optimization model is utilized for optimal portfolio selection. The effectiveness of this approach is demonstrated through comprehensive comparisons with alternative machine learning models, including support vector machines (SVM), LS-SVM, ARIMA, combined ARIMA+SVM models, and several benchmarking models from the existing literature. Validation of the proposed technique is conducted using historical data from the Bombay Stock Exchange (BSE), India.

1. Introduction

Portfolio selection is a decision-making process in financial mathematics that focuses on allocating the investor's wealth to a set of financial assets while the allocating weight is constantly changed to maximize the profit and minimize the associated risk. The portfolio theory presented by Markowitz [1] serves as a crucial foundation for portfolio management, a field that has been extensively researched but has not yet been entirely conquered. Constructing a portfolio entails two key challenges. One involves selecting assets with higher potential returns to minimize risk, while the other revolves around assessing the contribution of each asset to the portfolio's overall value. Markowitz's mean-variance (MV) model is the foundation for this research, and various studies have added numerous sensible insights concerning portfolio construction [2–8]. However, the average return on an asset is the key aspect of the portfolio optimization process, which makes the pre-selection of assets an essential process for portfolio optimization. Asset selection has been a significant but challenging subject in the area of financial investment. This field of study depends on the performance of financial time-series data in the past as well as the accurate performance predictions of assets for the future [9]. Moreover, the stock price is a time series data made up of numerous independent variables. It is very unstable and complex with a high degree of nonlinearity and time variation [10]. Therefore, enhanced forecasting methodologies are required for stock forecasting.

Due to the ineffectiveness of conventional approaches in handling intricate, multi-dimensional, and noisy time-series data, there has been a growing trend among economists in recent years to incorporate machine learning into the process of forecasting stock prices [11–16]. By examining existing research, it becomes evident that while accurately forecasting stock market patterns is challenging, it is possible to predict future trends to some degree, thereby mitigating risks for investors. Currently, despite the availability of many approaches to forecast stock market prices, scholars from both domestic and international arenas have adopted diverse financial time series models to improve their understanding and control of fluctuations in financial markets. The Generalized AutoRegressive Conditional Heteroskedasticity (GARCH) model and the ARIMA model are two of the most popular and commonly used. With time series data being generated over time and the growing dependence of financial market research and analysis on such data, recently, many researchers [17–22] have incorporated pre-selection of assets using ARIMA with portfolio optimization. For example, Lin [23] displays a fresh attempt to use the ARIMA model and back propagation neural network algorithm to predict the daily price of bitcoin. The bitcoin portfolio problem is further treated using a modified Markowitz portfolio investing model. In Mihova et al. [24], the asset prices of four Bulgarian enterprises are forecasted using ARIMA models, and then rates of return are computed from the models. Accordingly, an ideal risk portfolio has been created based on the

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Markowitz model.

Despite the widespread popularity of the ARIMA model, the decision-making process for selecting an appropriate forecasting model involves multiple factors, with accuracy being recognized as a crucial criterion. Consequently, enhancing forecasting accuracy has become imperative for decision-makers and managers across various scientific domains, particularly in the realm of time series forecasting. Many researchers advocate that the effectiveness of forecasting accuracy can be significantly improved by either combining different models or employing hybrid models, thereby addressing the limitations of individual models. Both theoretical and empirical evidence in the literature suggest that hybrid models, formed by combining diverse models, exhibit lower generalization variance or error. Generally, the concept of combination usually arises from the inherent difficulty in determining the underlying process of data generation. In other words, it is challenging for a single model to accurately identify the true data generation process or capture all the characteristics of a time series entirely [25]. Furthermore, the ARIMA model operates under the assumption that the current value is a linear combination of past values and past errors. The requirement of stationarity is essential when applying an ARIMA model for forecasting. Therefore, being a linear model, ARIMA is solely applicable for constructing models that capture linear patterns within time series data. Over the past few decades, the combination of various forecasting models has garnered significant interest among researchers, particularly in the field of financial time series forecasting. Several researchers have combined different models with the ARIMA model for stock market forecasting in order to overcome the limitations of a single prediction model and improve the accuracy [26–29]. Alam et al. [30] have proposed an ARIMA-LSTM hybrid model to forecast the correlation coefficients of assets in the Dhaka Stock Exchange (DSE) market, which is essential for portfolio optimization. The outcomes of their experiments reveal that the hybrid model surpasses alternative models in terms of accuracy and suggests that the ARIMA-LSTM hybrid model represents an effective approach for predicting correlation coefficients crucial for portfolio optimization. The ARIMA-LSTM model is employed by Hong [31] for forecasting the closing price of stocks for the next 10 working days. Following this, a mean–variance model is constructed, and the optimal portfolio is determined based on maximizing the Sharpe ratio. The findings indicate that the integrated ARIMA-LSTM model demonstrates superior accuracy compared to individual ARIMA and LSTM models. Moreover, utilizing the predicted stock closing prices from the combined model, the performance of the maximum Sharpe ratio portfolio outperforms the market average. Nevertheless, there is still a scarcity of studies focusing on portfolio optimization using the hybrid ARIMA model.

In addition to that, SVM is a nonlinear model rooted in statistical learning theory and the principle of structural risk minimization, designed to address the overfitting issue to a certain extent. SVM has found widespread application in time series forecasting across various domains, including tax revenue forecasting, electric power consumption forecasting, and stock market trend forecasting [32]. In 1999, Suykens & Vandewalle [33] introduced the Least Squares Support Vector Machine (LS-SVM), which enhances the speed of solving optimization problems compared to the conventional SVM model. In recent years, LS-SVM has been used in many research fields and has shown noteworthy results [34–43]. Moreover, by integrating LS-SVM with other techniques, researchers have obtained remarkable results, particularly in the realm of financial forecasting [10,44–47]. In Zhao et al. [48], the least squares support vector machine is utilized for forecasting systemic financial risk and the findings indicate that LS-SVM is better suited for practical applications in predicting systemic financial risk. An enhanced particle swarm optimization (IPSO) algorithm is introduced by Gong et al. [49] to optimize the parameters of the LS-SVM technique for predicting stock returns' volatility. The study compares forecasting performances of AGARCH-nG models, hybrid AGARCH-nG-ANN methods, and the LS-SVM-IPSO approach using

stock market data. Empirical results confirm the LS-SVM-IPSO approach's effectiveness and superiority, outperforming AGARCH-type models with non-Gaussian distributions and those integrating artificial neural network methods. A growing body of research has explored both linear and non-linear models to improve the accuracy of financial forecasts, highlighting the distinct advantages and limitations of each approach.

2. Prior works

Stock prediction and portfolio optimization are considered crucial in financial decision-making. Accurate stock price forecasting is used to balance profits and risks, aiding resource allocation as efficiently as possible. Over the years, a variety of linear and non-linear models have been employed in financial time series prediction. Among these, ARIMA, widely recognized as the linear Box–Jenkins model, is extensively used due to its straightforward application and reliable results. Various time series, including stock index returns and exchange rates, have been foretasted using the ARIMA model. Comparisons with models such as GARCH, Markov switching, and regime switching have shown that ARIMA models often outperform others [50]. Its widespread use has established the ARIMA model as a benchmark for evaluating novel modeling techniques [51].

While the ARIMA model is regarded as reliable for linear datasets, its limitations in handling erratic, nonlinear data have been identified, prompting the development of hybrid models. These models have been created by integrating ARIMA with advanced machine-learning methods to address its shortcomings in modeling nonlinear patterns, thereby enhancing forecasting accuracy. Neural network models such as artificial neural networks (ANN) [52–55], recurrent neural networks (RNN) [56], multilayer perceptrons (MLP) [25,57,58], long short-term memory (LSTM) [59–65], and Gated recurrent unit (GRU) [66] have been employed to capture nonlinear stock market movements. In these studies, autoregressive terms have been used as neural network inputs, while residuals from ARIMA models have been fed into neural network models to form hybrid ARIMA-neural network models. The performance of these hybrid models has been compared with standalone ARIMA and neural network models. Experimental findings using real datasets have demonstrated the hybrid model's potential to improve prediction accuracy. However, while neural networks are recognized for their efficiency in time series modeling and forecasting, their limitations in pattern recognition reduce their suitability for financial applications requiring time-sensitive decisions.

Hybrid models such as ARIMA-GARCH [11,67–70] and ARIMA-SVM [26,71] have been widely employed for time series forecasting. SVM has gained popularity due to its superior generalization capabilities and stable parameterization, regardless of input dimension size. Research has shown that SVM outperforms neural networks in financial time series forecasting, with hybrid models delivering enhanced accuracy and profitability. Despite ARIMA+GARCH's effectiveness in modeling volatility, it lacks the ability to capture complex price dynamics. Similarly, while ARIMA+SVM is adept at handling nonlinearity, it is constrained by overfitting and computational inefficiencies. To address these limitations, LS-SVM has been increasingly adopted for time series forecasting [72]. This approach, used in both prediction and classification tasks, has advanced hybrid modeling, particularly in ARIMA+LS-SVM frameworks. Yan and Chowdhury [73] demonstrated its application by integrating ARIMA with external input (ARMAX) and LS-SVM for energy market forecasting. Sun and Zhang [74] further enhanced LS-SVM-based carbon price prediction using the artificial fish swarm algorithm (AFSA), with experimental results identifying ARIMA+LS-SVM as the most accurate and computationally efficient hybrid model. Its scalability, adaptability, and ability to balance performance and efficiency make it a preferred choice for stock price forecasting and portfolio optimization tasks requiring timely and precise predictions.

The combined ARIMA model leverages complementary features to improve stock forecasts in asset allocation optimization. By integrating ARIMA's linear trend detection with models capturing non-linear dynamics, hybrid frameworks enhance portfolio optimization accuracy and have been shown to improve capital allocation and returns [30, 75,76]. While such models capture complex patterns, they often face challenges like interpretability, computational cost, and overfitting. In contrast, the ARIMA+LS-SVM hybrid balances performance, efficiency, and flexibility by modeling both linear and non-linear behaviors effectively. Its scalability and accuracy make it a reliable tool for portfolio optimization and a valuable choice for managing dynamic market data.

Research gap, motivation and objective

Despite extensive research in portfolio optimization using ARIMA and LS-SVM methodologies, their integration remains underexplored. While ARIMA excels in capturing linear patterns and LS-SVM is effective for nonlinear relationships, their combined application in portfolio optimization is absent in the literature. Furthermore, the following points are also notable.

- The literature predominantly focuses on either ARIMA-based or LS-SVM-based approaches, neglecting integration. This gap limits the exploration of potential synergies that could enhance forecasting accuracy and risk management in portfolio optimization.
- Both ARIMA and LS-SVM models have limitations: ARIMA assumes linear patterns and requires stationarity, while LS-SVM is nonlinear and susceptible to overfitting. Combining these models could mitigate these limitations and establish a more robust forecasting framework.

The primary motivation is to enhance forecasting accuracy in portfolio optimization, thereby improving decision-making for investors. Leveraging the strengths of ARIMA and LS-SVM, the hybrid approach is expected to provide more accurate stock price predictions, while also addressing the dynamic nature of financial markets. By capturing both linear and nonlinear patterns, the hybrid ARIMA+LS-SVM approach enhances forecasting accuracy, adaptability to market changes, and risk management capabilities. These advantages can collectively result in more effective and efficient portfolio optimization strategies. This serves as the inspiration for the authors to investigate the hybrid ARIMA+LS-SVM approach in portfolio optimization. The primary contributions of this study include:

- A combined ARIMA+LS-SVM approach is introduced to develop a portfolio optimization algorithm, which combines auto-regressive integrated moving average with the least-square support vector machine.
- The ARIMA+LS-SVM model is utilized for stock price prediction, and stocks demonstrating superior performance are chosen for the portfolio optimization process.
- The performance of the ARIMA+LS-SVM algorithm is examined by obtaining the efficient solution for the mean-variance portfolio optimization model.
- To assess the effectiveness of the proposed algorithm, the model is compared with various existing models like the ARIMA model, SVM model, LS-SVM model, and hybrid ARIMA+SVM model and with some of the benchmark portfolio optimization models from the literature.
- To evaluate the applicability of the suggested methodology, an experimental analysis is conducted utilizing a historical dataset sourced from the Bombay Stock Exchange, India.

This study contributes to advancing hybrid forecasting methodologies and their application in portfolio optimization, offering a novel approach to balancing accuracy, efficiency, and adaptability in financial

decision-making.

The remainder of the paper follows this structure: Section 3 presents the mean-variance portfolio optimization model. Section 4 elaborates on the experimental process including data and the forecasting methods employed within it. The findings of the empirical studies are deliberated upon in Section 5. Lastly, Section 6 wraps up with a summary of the main discoveries and a discussion on future prospects.

3. Mean-variance portfolio optimization model

Consider an investor aiming to allocate funds to n (where $n \geq 2$) risky assets for a fixed time period T . Let r_j represent the rate of return for asset j . The expected rate of return for asset j , denoted as μ_j , is calculated as $\mu_j = \frac{1}{T} \sum_{t=1}^T r_{jt}$. Assume x_j is the proportion of wealth invested in asset j , subject to the constraint $\sum_{j=1}^n x_j = 1$. Consequently, the rate of return for the portfolio (r_p) is expressed as $r_p = \sum_{j=1}^n r_j x_j$, the expected return for the portfolio (μ_p) is $\mu_p = \sum_{j=1}^n \mu_j x_j$, and the portfolio's variance (σ_p^2) is given by $\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} x_i x_j$.

The Markowitz mean-variance model is primarily designed to ascertain the allocation of investments among individual stocks, denoted by $j = 1, 2, \dots, n$, within a specified period. This model utilizes mean and variance derived from historical asset prices to quantify the anticipated return and risk associated with the resulting portfolio. The underlying assumption of the MV is that the investor aims to either maximize the expected return at a given level of risk or minimize the risk for a specified level of return. The Markowitz's MV model is described as follows:

$$\min \sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} x_i x_j \quad (1)$$

$$\text{subject to: } \sum_{j=1}^n \mu_j x_j = \mu_{fix}, \quad (2)$$

$$\sum_{j=1}^n x_j = 1, \quad (3)$$

$$0 \leq x_j < 1, \forall j = 1, 2, \dots, n. \quad (4)$$

In the above-described model, the goal of the objective function (1) is to minimize the risk, which is the variance of the stock portfolio, by adjusting the allocation ratios of investments in shares of listed companies. The portfolio's expected return must reach the predetermined minimum expected return value μ_{fix} , which is stated in constraint (2). Constraint (3) ensures that the total investment in shares of the listed companies does not surpass one. It also stipulates in constraint (4) that the investment ratios should not be negative.

4. Experimental process

4.1. Data

This paper utilizes historical data from the BSE 100 component stocks as the experimental dataset. The BSE 100 index comprises the top 100 constituent stocks, offering a comprehensive reflection of the overall performance of companies with significant market value and market impact in the Indian stock market. The stocks under consideration are listed in Table 1, providing their respective stock codes along with average return and variance values. The historical data spans from January 1, 2010, to June 30, 2023, encompassing the COVID-19 event period. Therefore, training the model on stock data that includes such events, which may lead to fluctuations in stock prices, adds more interest. Following data collection, data preprocessing is conducted, which includes data cleaning to address any missing values. Daily closing prices for each stock serve as input features for the models. The dataset is split into a training set and a testing set with a ratio of 8:2. The model is trained using the training set, while the testing dataset

Table 1

List of stock numbers and their corresponding stock codes.

Stock no	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Code	500 034	500 049	500 087	500 096	500 112	500 114	500 124	500 180	500 182	500 209
Expected return	0.001583	0.000078	0.000458	0.000546	0.000219	0.001034	0.000584	0.000405	0.000317	0.000121
Variance	0.000759	0.000965	0.000258	0.000295	0.000684	0.000696	0.000269	0.000481	0.000319	0.000520
Stock no	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20
Code	500 228	500 247	500 251	500 257	500 295	500 300	500 312	500 325	500 331	500 387
Expected return	0.000625	0.000506	0.000836	0.000261	0.000263	0.000292	−0.000168	0.000468	0.000969	0.000913
Variance	0.000808	0.000453	0.000687	0.000517	0.000760	0.000526	0.000619	0.000379	0.000340	0.000334
Stock no	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30
Code	500 400	500 425	500 440	500 469	500 470	500 477	500 480	500 490	500 493	500 510
Expected return	0.000101	0.000608	0.000605	0.000367	0.000157	0.000720	0.000639	0.000876	0.000632	0.000313
Variance	0.000694	0.000384	0.000645	0.000783	0.000776	0.000707	0.000367	0.000323	0.000560	0.000382
Stock no	S31	S32	S33	S34	S35	S36	S37	S38	S39	S40
Code	500 520	500 547	500 550	500 570	500 696	500 790	500 800	500 820	500 825	500 830
Expected return	0.000370	0.000202	0.000737	0.000487	0.000799	0.000760	0.000583	0.000736	0.000723	0.000424
Variance	0.000507	0.000652	0.000360	0.000867	0.000217	0.000209	0.000637	0.000498	0.000509	0.000266
Stock no	S41	S42	S43	S44	S45	S46	S47	S48	S49	S50
Code	500 850	500 875	503 806	505 200	507 685	508 869	511 218	511 243	512 070	512 599
Expected return	0.000632	0.000373	0.001219	0.001135	0.000062	0.000872	0.000734	0.001419	0.000680	0.001300
Variance	0.000487	0.000350	0.000714	0.000680	0.000407	0.000513	0.000695	0.000854	0.000535	0.001247
Stock no	S51	S52	S53	S54	S55	S56	S57	S58	S59	S60
Code	517 354	523 642	524 715	530 965	531 642	532 134	532 155	532 174	532 187	532 215
Expected return	0.000743	0.001384	0.000352	−0.000048	0.000685	0.000269	−0.000125	0.000472	0.000986	0.000484
Variance	0.000669	0.000882	0.000590	0.000549	0.000341	0.000845	0.000510	0.000619	0.000625	0.000673
Stock no	S61	S62	S63	S64	S65	S66	S67	S68	S69	S70
Code	532 281	532 343	532 424	532 432	532 354	532 461	532 483	532 488	532 500	532 538
Expected return	0.000624	0.001205	0.000679	0.000415	0.000494	−0.000308	0.000275	0.000736	0.000715	0.000788
Variance	0.000467	0.000627	0.000484	0.000759	0.000397	0.000844	0.000723	0.000428	0.000330	0.000290
Stock no	S71	S72	S73	S74	S75	S76	S77	S78	S79	S80
Code	532 540	532 555	532 648	532 755	532 777	532 810	532 868	532 898	532 921	532 955
Expected return	0.000624	0.000073	0.000065	0.000407	0.000858	0.000322	0.000507	0.000368	0.000617	0.000239
Variance	0.000322	0.000267	0.001503	0.000535	0.000672	0.000700	0.000830	0.000242	0.000766	0.000675
Stock no	S81	S82	S83	S84	S85	S86	S87	S88	S89	S90
Code	532 977	532 978	533 179	533 278	533 758	539 437	539 448	540 005	540 180	540 376
Expected return	0.000486	0.001084	0.001027	0.000047	0.001423	0.000224	0.000502	0.000716	0.000402	0.000641
Variance	0.000335	0.000727	0.000500	0.000330	0.000951	0.000348	0.000349	0.000233	0.000416	0.000205
Stock no	S91	S92	S93	S94	S95	S96	S97	S98	S99	S100
Code	540 611	540 716	540 719	540 762	540 777	541 154	541 729	542 830	543 220	543 320
Expected return	0.000308	0.000285	0.000255	0.000870	0.000275	0.000468	0.000148	0.000276	0.000566	−0.000057
Variance	0.000373	0.000164	0.000143	0.000261	0.000168	0.000215	0.000155	0.000379	0.000134	0.000196

is employed to assess the performance of the applied model. In the experiment, SVM, LS-SVM, ARIMA, ARIMA+SVM, and ARIMA+LS-SVM models are prepared using the Scikit-learn machine-learning package in Python software.

4.2. Prediction models

4.2.1. Autoregressive integrated moving average

The ARIMA model, pioneered by statisticians George Box and Gwilym Jenkins [77], is widely recognized as a prominent methodology for time series analysis. It has been a predominant approach in the realm of time series forecasting across various domains. In a given time series x_t , $t = 1, 2, \dots, n$, the ARIMA model can be formulated as given below.

$$\phi(B)(1 - B)^d x_t = \theta(B)a_t, \quad (5)$$

$$\phi(B) = 1 - \phi_1(B) - \phi_2(B)^2 - \dots - \phi_p(B)^p, \quad (6)$$

$$\theta(B) = 1 - \theta_1(B) - \theta_2(B)^2 - \dots - \theta_q(B)^q, \quad (7)$$

where $\phi(B)$ represents the autoregressive (AR) operator with an order of p , and $\theta(B)$ represents the moving average operator with an order of q . The integers p , d , and q serve as parameters, with d , indicating the order of autoregressive terms, p denoting the number of nonseasonal differences, and q representing the degree of moving average terms. The ARIMA model is fundamentally a data-driven methodology derived from the inherent structure of the data. Its fundamental principle lies in the assumption that there is a linear correlation structure among the

values in the time series [78].

In ARIMA (p, q, d) , first, a reasonable range for p and q is chosen. Additionally, the Augmented Dickey–Fuller (ADF) test is used to check the stationarity, and the parameter d is set based on whether differencing is needed. The considered values for the hyperparameters are presented in Table 2. Further, the grid search method is implemented to obtain the best (p, q, d) values. After multiple attempts, the optimal hyperparameters for the ARIMA have been established, and the considered hyperparameters are (p, d, q) as $(0, 1, 0)$.

4.2.2. Support vector machine

In order to address binary classification issues, Support Vector Machines (SVM) were created as an approach. It was also expanded to solve other issues, like regression [79]. Due to its outstanding performance, SVM is now a standard in many machine-learning programs for both regression and classification. By employing convex optimization approaches, Support Vector Regression (SVR) simplifies difficult nonlinear regression issues into more manageable linear ones. This transformation allows for a fine-tuning of the relationship between model complexity and empirical error, leading to very accurate predictions. To represent the non-linear connection between the random variables X and Y in SVR, the following linear function f is used:

$$f(x) = w\phi(x) + b,$$

where $f(x)$ represents the simulated value, and the parameters $w \in \mathbb{R}^n$ and $b \in \mathbb{R}$ can be changed [80]. SVR is formulated based on minimizing structural and empirical risks as presented in Eq. (8) [81].

Table 2
Parameters of the prediction models.

Model	Parameter	Values
ARIMA	p	0, 1, 2, 3, 4, 5
	q	0, 1, 2, 3, 4, 5
	d	0, 1, 2
SVM	c	0.1, 1, 10, 100
	γ	0.001, 0.01, 0.1, 1
	ϵ	0.01, 0.1, 0.5, 1
LS-SVM	γ	0.1, 1, 10, 100
	σ^2	0.001, 0.01, 0.1, 1

$$\min R(w, \xi, \xi^*) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i^* + \xi_i) \quad (8)$$

$$\text{subject to: } y_i - w\phi(x_i) - b \leq \epsilon + \xi_i, \quad i = 1, 2, \dots, n, \quad (9)$$

$$-y_i + w\phi(x_i) + b \leq \epsilon + \xi_i^*, \quad i = 1, 2, \dots, n, \quad (10)$$

$$\xi_i^* \geq 0, \quad i = 1, 2, \dots, n, \quad (11)$$

$$\xi_i \geq 0, \quad i = 1, 2, \dots, n. \quad (12)$$

This work uses the radial basis function as the kernel function, which is given as follows

$$k(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2).$$

Both kernel co-efficient (γ) and the regularization parameter (C) comprise the parameters of SVR. These parameters' utilized values are shown in Table 2. By utilizing the grid search method, this paper sets the values for C , γ , and ϵ as 0.1, 0.001, and 0.1.

4.2.3. Least squares support vector machine

Suykens et al. [82] introduced the concept of LS-SVM models as an alternative expression of SVM regression. LS-SVM offers comparable benefits to SVM, and notably, it involves solving a system of linear equations (linear programming) rather than a quadratic programming (QP) problem. This computational simplification enhances the ease of handling the problem. Consider a provided training set comprising N data points, $\{x_k, y_k\}_{k=1}^N$, where the input data x_k belongs to the N -dimensional vector space R^N , and the output y_k belongs to the one-dimensional vector space r . In feature space, the LS-SVM model is represented as,

$$y(x) = w^T \phi(x) + b, \quad (13)$$

where the input data undergoes a transformation into a higher-dimensional feature space through the feature map $\phi(\cdot)$, w serving as an adjustable weight vector in the space R^N , and b is a scalar threshold in the space r . In the context of Least Squares Support Vector Machines (LS-SVM), the optimization problem for function estimation is explicitly defined as follows.

$$\min \frac{1}{2} w^T w + \frac{\gamma}{2} \sum_{k=1}^N e_k^2 \quad (14)$$

$$\text{subject to: } y(x) = w^T \phi(x_k) + b + e_k, \quad k = 1, 2, \dots, N, \quad (15)$$

where N represents the number of data and γ is the regularization parameter. After solving the above optimization problem, it is obtained that,

$$y(x) = \sum_{k=1}^N \alpha_k K(x, x_k) + b, \quad (16)$$

where the radial basis function $K(x_k, x_l) = \exp\{-\frac{\|x_k - x_l\|^2}{2\sigma^2}\}$, $k = 1, 2, \dots, N$ and σ is refers to the radial basis function's width. Furthermore, the best values for the LS-SVM model's hyperparameters have been found using the grid search technique; these values are regularization parameter $\gamma = 1$ and kernel co-efficient $\sigma^2 = 0.001$. Apart

from this, the methodological framework for portfolio optimization using single models like SVM, LS-SVM, and ARIMA is illustrated in Fig. 1.

4.2.4. The hybrid ARIMA+SVM

It is difficult to record the movement of the price of shares. For stock price forecasting, a hybrid approach that combines both nonlinear and linear modeling capabilities is a reasonable substitute. The ARIMA and SVM models possess distinct abilities to detect data properties in linear and nonlinear realms; hence, the combination of these models can effectively represent linear as well as nonlinear patterns, enhancing overall predicting performance [83]. The following represents the hybrid model f_t :

$$f_t = \hat{z}_t + \gamma_t, \quad (17)$$

where the ARIMA model's output at time t is represented by $\hat{z}_t$, and the residual at time t is indicated by γ_t . In its residuals, the ARIMA model was unable to identify nonlinear patterns. Thus, SVM is used to model the residuals in the next step, which makes it possible to identify nonlinear connections. The following is a representation of the residuals modeled by SVM:

$$\gamma_t = f(\gamma_{t-1}, \gamma_{t-2}, \dots, \gamma_{t-n}) + \Delta_t, \quad (18)$$

where the random error is represented by Δ_t and the nonlinear function that the SVM model is f . Suppose, the prediction for the residual given in (18) is represented by $\hat{\gamma}_t$. Consequently, the final combined projected model will be

$$f_t = \hat{z}_t + \hat{\gamma}_t. \quad (19)$$

Furthermore, the best hyperparameters for the ARIMA and SVM models have been determined by the grid search approach. These include $(p, d, q) = (0, 1, 0)$ and $(C, \gamma, \epsilon) = (0.1, 0.001, 0.1)$.

4.2.5. The hybrid ARIMA+LS-SVM

Capturing the behavior of stock prices is challenging. Due to the intricate nature of investor behavior, leading stock indices show both linear and nonlinear patterns. Single models often underperform in forecasting high-complexity time series data. ARIMA struggles to grasp nonlinear information as it inherently relies on linear characteristics. Additionally, time series data evolve, continuously updating with the latest historical information, making it uncertain if the series consistently aligns with model requirements. Both ARIMA and LS-SVM individually fall short in accurately forecasting stock indices due to their inability to model both linear and nonlinear patterns [84]. Therefore, the use of hybrid models in time series forecasting aims to mitigate the risk of model specification errors.

In the hybrid model, the LS-SVM component is incorporated to capture residual information that the ARIMA model might not have learned. Additionally, the forecasted values generated by LS-SVM signify the non-linear information that extends beyond the learning capacity of the ARIMA model. The combination of ARIMA and LS-SVM can be expressed as follows:

$$y_t = \hat{f}_t + \epsilon_t \quad (20)$$

$$\Rightarrow \epsilon_t = y_t - \hat{f}_t, \quad (21)$$

where $\hat{f}_t$ is the output of the ARIMA model at time t and ϵ_t denotes the residual at time t of the ARIMA model. The ARIMA model failed to capture nonlinear patterns within its residuals. Consequently, in the subsequent stage, the residuals are modeled using LS-SVM, enabling the identification of nonlinear relationships. The LS-SVM model for the residuals, characterized by n input nodes, is formulated as follows:

$$\hat{\epsilon}_t = f(\epsilon_{t-1}, \epsilon_{t-2}, \dots, \epsilon_{t-n}) + e_t, \quad (22)$$

where f represents a nonlinear function determined by the LS-SVM, and e_t denotes the random error. Consider $\hat{\epsilon}_t$ represents the forecast of

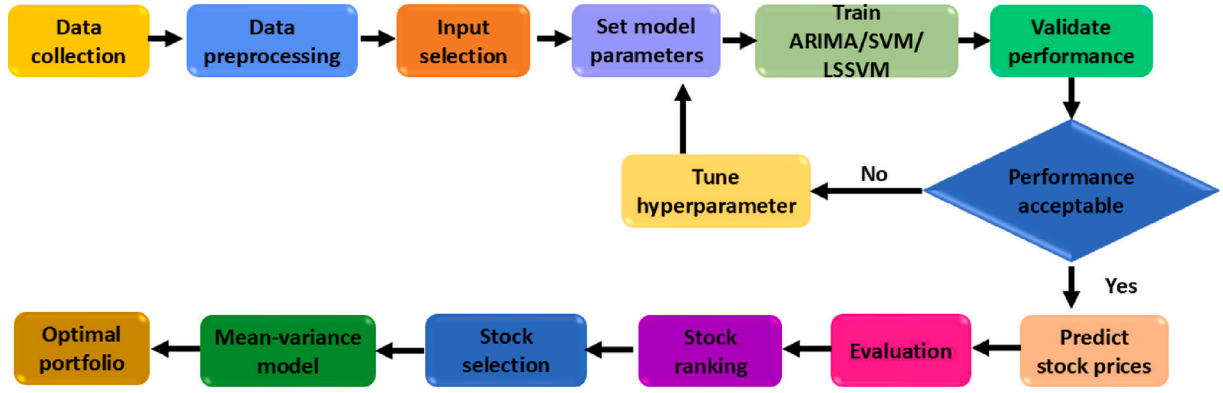


Fig. 1. The methodological framework to portfolio optimization using SVM, LS-SVM, and ARIMA.

the residual provided in (22). Therefore, the final combined forecasted model will be

$$y_t = \hat{f}_t + \hat{\epsilon}_t. \quad (23)$$

Additionally, in this study, using the grid search method, the optimal hyperparameters for the ARIMA and LS-SVM model have been established, and the considered hyperparameters are $(p, d, q) = (0, 1, 0)$, and $(\gamma, \sigma^2) = (1, 0.001)$.

4.2.6. Methodology

In the financial market, individual investors commonly seek guidance on determining the most effective strategy for building an optimal investment portfolio. The goal is to create a portfolio that maximizes potential returns while minimizing risks. Therefore, the focus primarily revolves around selecting better-performing stocks to construct the portfolio from a predictive standpoint. However, certain factors, such as economic crises and economic growth, exert a substantial influence on stock prices in financial markets. Consequently, there is a necessity to investigate the nonlinear component of stock prices in time series to enhance forecasting accuracy. Thus, this study suggests a hybrid approach that integrates ARIMA and LS-SVM for forecasting stock prices and employs high-quality stocks in the portfolio optimization model to obtain efficient investment strategies.

A general approach for time series analysis involves decomposing the series into three components: trend, seasonality, and the residual component represents the random variation. This method offers a structured framework for addressing forecasting challenges within a time series, considering both the intricacies of modeling and the optimal capture of these components in a given model. In this section, the time series of stock prices undergoes a division into two components using the ARIMA model: the trend component and the irregular (residual) component. The hybrid approach proposed is outlined in Fig. 2, and the steps are explained as follows:

- **Step 1:** The ARIMA method is employed to fit the trend component (closing price of the stock) of the series and generate a series of forecasts ($\hat{f}_t$).
- **Step 2:** A sequence of irregular elements (ϵ_t) is derived by deducting the predictions of the trend component from the actual stock values (y_t).

$$\epsilon_t = y_t - \hat{f}_t \quad (24)$$

- **Step 3:** The LS-SVM method is employed to fit the irregular component (ϵ_t) and produce a series of forecasts ($\hat{\epsilon}_t$).

Table 3
List of evaluation metrics and their formulas.

Metric	Formula
MSE	$MSE = \frac{1}{N} \sum_{j=1}^N (y_j - \hat{y}_j)^2$
RMSE	$RMSE = \sqrt{\frac{1}{N} \sum_{j=1}^N (y_j - \hat{y}_j)^2}$
MAE	$MAE = \frac{1}{N} \sum_{j=1}^N y_j - \hat{y}_j $

- **Step 4:** The ultimate predictions (y_t) for stock prices are computed, as depicted in the following equation.

$$y_t = \hat{f}_t + \hat{\epsilon}_t \quad (25)$$

- **Step 5:** Subsequently, stocks demonstrating superior performance and minimal prediction error as per the model are prioritized for the portfolio optimization procedure.
- **Step 6:** Finally, the mean-variance portfolio optimization model integrates the chosen high-performing stocks to achieve the most efficient portfolio.

5. Experimental results

5.1. Performance of stock prediction process

The aim of predicting stock price is to identify stocks with better performance and earning potential for portfolio optimization. Assigning varying weights to these stocks allows for the possibility of achieving high returns with low risk. Consequently, the selection of an appropriate evaluation index for forecasting becomes crucial. In this section, the effectiveness of various models in predicting stock prices is assessed using three metrics: mean absolute error (MAE), mean squared error (MSE), and root mean square error (RMSE) [85]. The formulas for the evaluation metrics are presented in Table 3, where y_j and $\hat{y}_j$ represent the actual value and predicted value at time t respectively. Moreover, the prediction performance of the ARIMA+LS-SVM model is evaluated against alternative models, including the ARIMA model, SVM model, LS-SVM model, and hybrid ARIMA+SVM model.

The performance of prediction models, evaluated using MAE, MSE, RMSE, and computation time (in seconds), is summarized in Table 4. Among single models, LS-SVM achieves the lowest errors (MAE: 0.000122, MSE: 0.686276, RMSE: 0.000129) with a relatively fast computation time of 22.417 s, making it more accurate and efficient than

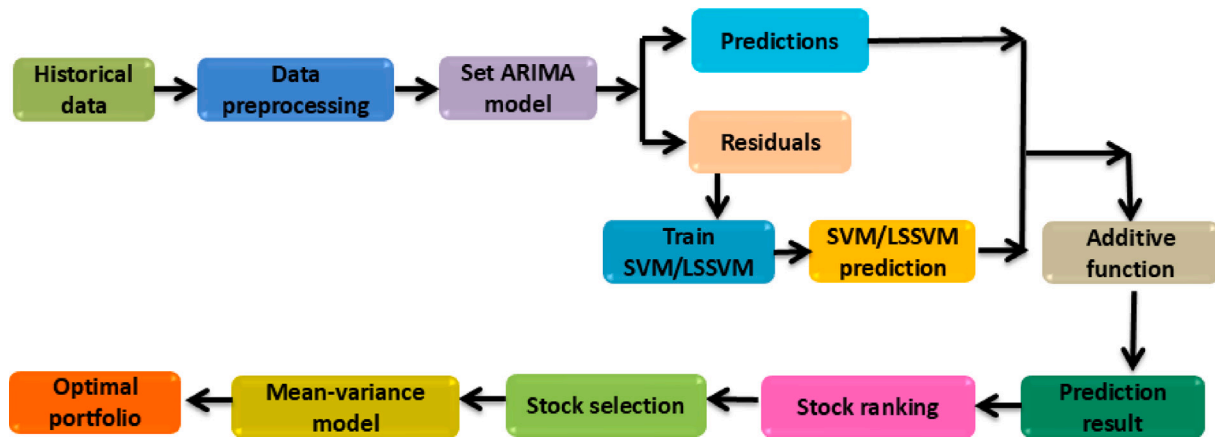


Fig. 2. The methodological framework to portfolio optimization using ARIMA+SVM and ARIMA+LS-SVM.

Table 4

Performance assessment of various models in prediction.

Model		MAE	MSE	RMSE	Time (s)
SVM	Mean	0.001256	0.800960	0.001338	88.221
	Std	0.002416	3.087503	0.002507	
LS-SVM	Mean	0.000122	0.686276	0.000129	22.417
	Std	0.000222	2.616905	0.000229	
ARIMA	Mean	0.000750	1.028893	0.001497	16.740
	Std	0.001322	4.140065	0.002346	
ARIMA+SVM	Mean	0.001329	0.884505	0.001405	94.546
	Std	0.002435	1.433664	0.002523	
ARIMA+LS-SVM	Mean	0.000106	0.051044	0.000113	30.890
	Std	0.000294	0.305830	0.000298	

both SVM and ARIMA. ARIMA, the fastest single model with a computation time of 16.74 s, exhibits higher errors compared to LS-SVM. For hybrid models, ARIMA+LS-SVM demonstrates the best performance, achieving the lowest errors (MAE: 0.000106, MSE: 0.051044, RMSE: 0.000113) with a computation time of 30.89 s, effectively balancing accuracy and efficiency. In contrast, ARIMA+SVM shows higher errors (MAE: 0.001329, MSE: 0.884505, RMSE: 0.001405) and is the slowest model, requiring 94.546 s. Overall, ARIMA+LS-SVM offers the best trade-off between accuracy and computation time, significantly enhancing accuracy while maintaining moderate computational demands. For scenarios prioritizing accuracy with minimal computation, LS-SVM emerges as the most suitable single model.

Building on the previous analysis of the ARIMA+LS-SVM model, Table 5 presents the results for various (p, d, q) configurations, evaluated based on MAE, MSE, RMSE, and computation time. The $(0, 1, 0)$ configuration demonstrates the best performance with the lowest errors (MAE: 0.000106, MSE: 0.051044, RMSE: 0.000113) and efficient computation time (30.89 s). Slightly higher errors are observed for $(1, 1, 0)$, $(1, 1, 1)$, and $(1, 1, 2)$, with $(1, 1, 1)$ taking longer (64.034 s). Configurations $(2, 1, 1)$ and $(2, 1, 2)$ have similar errors to $(1, 1, 2)$ but significantly higher computation times (84.772 and 162.089 s). The $(4, 1, 2)$ configuration achieves comparable accuracy (MAE: 0.000107) but requires the highest computation time (221.361 s). The $(5, 2, 0)$ configuration has the largest errors (MAE: 0.000139, MSE: 0.131989, RMSE: 0.000148) and moderate computation time (66.304 s). Overall, the $(0, 1, 0)$ configuration offers the best trade-off between accuracy and efficiency, highlighting its suitability for balancing prediction performance and computational demands.

5.2. Statistical test

In this section, two non-parametric statistical tests namely the Friedman and Nemenyi tests are employed to evaluate the forecasting accuracy (MAE, MSE, and RMSE) of prediction models across multiple stock

Table 5

Performance assessment of ARIMA+LS-SVM model at different (p, d, q) values.

(p, d, q) values		MAE	MSE	RMSE	Time (s)
$(0, 1, 0)$	Mean	0.000106	0.051044	0.000113	30.890
	Std	0.000294	0.305830	0.000298	
$(1, 1, 0)$	Mean	0.000110	0.088759	0.000116	33.474
	Std	0.000320	0.187223	0.000324	
$(1, 1, 1)$	Mean	0.000112	0.090481	0.000118	64.034
	Std	0.000330	0.187089	0.000333	
$(1, 1, 2)$	Mean	0.000108	0.087211	0.000115	84.276
	Std	0.000324	0.188150	0.000328	
$(2, 1, 1)$	Mean	0.000108	0.086934	0.000115	84.772
	Std	0.000324	0.187254	0.000328	
$(2, 1, 2)$	Mean	0.000109	0.088693	0.000115	162.089
	Std	0.000319	0.184776	0.000322	
$(4, 1, 2)$	Mean	0.000107	0.090520	0.000114	221.361
	Std	0.000299	0.191544	0.000303	
$(5, 2, 0)$	Mean	0.000139	0.131989	0.000148	66.304
	Std	0.000323	0.220939	0.000329	

datasets and to identify significant model pair differences. These tests are applied to SVM, LS-SVM, ARIMA, ARIMA+SVM, and ARIMA+LS-SVM, highlighting their strengths and limitations to determine the most effective approach for time series forecasting. The Friedman test is used to assess statistically significant differences in prediction accuracy across algorithms over several datasets. When differences are detected, the Nemenyi post hoc test identifies significant pairwise variations between models. The Scikit-learn library in Python is utilized for this analysis.

The Friedman non-parametric test is conducted under the null hypothesis (H_0) that the mean performance metrics of all models are identical. The alternative hypothesis (H_1) posits significant disparities in the models' average performances. The test results reveal substantial differences among the models for MAE (test statistic $\chi^2 = 294.0970485$, p -value = 2.03×10^{-62}), MSE (test statistic $\chi^2 = 201.397397$, p -value = 1.88×10^{-62}), and RMSE (test statistic $\chi^2 = 293.871999$, p -value = 2.27×10^{-62}). The large test statistics indicate considerable variability in the rankings of the models, while the extremely small p -values, far below common significance levels (e.g., 0.05 or 0.01), provide strong evidence against the null hypothesis. Consequently, the null hypothesis is rejected, concluding that at least one model's performance differs significantly from the others. Further, the Nemenyi post-hoc test is used to identify the particular models that differ from one another in a substantial way.

Pairwise comparisons of models (ARIMA, SVM, LS-SVM, ARIMA+SVM, and ARIMA+LS-SVM) using the Nemenyi test for MAE, MSE, and RMSE (Tables 6, 7, 8) reveal that ARIMA+LS-SVM consistently outperforms all others, with p -values of 0 against ARIMA, SVM, and ARIMA+SVM for MAE and significant differences across other metrics.

Table 6
Nemenyi post-hoc test p -values for MAE.

	ARIMA	SVM	LS-SVM	ARIMA+SVM	ARIMA+LS-SVM
ARIMA	1	0.650824	0	0.83145	0
SVM	0.650824	1	1.11E-16	0.111246	0
LS-SVM	0	1.11E-16	1	0	0.636464
ARIMA+SVM	0.83145	0.111246	0	1	0
ARIMA+LS-SVM	0	0	0.636464	0	1

Table 7
Nemenyi post-hoc test p -values for MSE.

	ARIMA	SVM	LS-SVM	ARIMA+SVM	ARIMA+LS-SVM
ARIMA	1	2.06E-08	0.914924	5.88E-15	0.000721
SVM	2.06E-08	1	7.79E-11	0.219129	0
LS-SVM	0.914924	7.79E-11	1	1.11E-16	0.016156
ARIMA+SVM	5.88E-15	0.219129	1.11E-16	1	0
ARIMA+LS-SVM	0.000721	0	0.016156	0	1

Table 8
Nemenyi post-hoc test p -values for RMSE.

	ARIMA	SVM	LS-SVM	ARIMA+SVM	ARIMA+LS-SVM
ARIMA	1	0.929185	0	0.991739	0
SVM	0.929185	1	0	0.720529	0
LS-SVM	0	0	1	0	0.578283
ARIMA+SVM	0.991739	0.720529	0	1	0
ARIMA+LS-SVM	0	0	0.578283	0	1

LS-SVM shows better performance than ARIMA and SVM (p -values 0 and 1.11×10^{-16}) but does not significantly differ from ARIMA+LS-SVM. ARIMA and ARIMA+SVM perform similarly ($p = 0.831450$) and are outperformed by ARIMA+LS-SVM. For MSE, ARIMA+LS-SVM remains superior, while ARIMA+SVM and SVM show no significant difference, and LS-SVM underperforms. In RMSE, ARIMA+LS-SVM is significantly better than ARIMA, SVM, and ARIMA+SVM but not LS-SVM (p -value = 0.578). Overall, ARIMA+LS-SVM is the most effective model, followed by LS-SVM, with ARIMA and SVM being the weakest and ARIMA+SVM showing moderate performance.

The process of pre-selecting stocks plays a crucial role before engaging in portfolio optimization, and an enhanced stock pre-selection can contribute to further improving the performance of portfolio management. Therefore, the selection of stocks with more accurately predicted stock prices is essential prior to portfolio optimization. After the completion of the prediction process, only 30 stocks out of the initial 50 from each model are included in the subsequent stage. This selection is determined by the stocks' performance, and those with higher RMSE values are excluded from the list. Table 9 showcases the 30 selected stocks from the five distinct models.

5.3. Performance of portfolio optimization model

This section presents the results of various experiments conducted with mean-variance portfolio optimization (MVPO) models. Moreover, the study employs SVM+MVPO, LS-SVM+MVPO, ARIMA+MVPO, ARIMA+SVM+MVPO, and ARIMA+LS-SVM+MVPO to represent MVPO models incorporating SVM-predicted stocks, LS-SVM-predicted stocks, ARIMA-predicted stocks, ARIMA+LS-SVM-predicted stocks, and ARIMA+LS-SVM-predicted stocks, respectively. The nonlinear programming problem-based MVPO model ((1)–(4)) has been assessed using the LINGO 11 optimization solver to determine the optimal investment proportions.

This section initiates the analysis of results by determining the optimal portfolio while varying the values of μ_{fix} , specifically at 0.0003, 0.0005, 0.0007, 0.0009, 0.0011, 0.0013, 0.0014, for the SVM+MVPO, LS-SVM+MVPO, ARIMA+MVPO, ARIMA+SVM+MVPO, and ARIMA+LS-SVM+MVPO models. Additionally, the variance, standard deviation, and Sharpe ratio (SR) values are computed for all portfolios associated with the SVM+MVPO, LS-SVM+MVPO, ARIMA+MVPO, ARIMA+SVM+

and ARIMA+LS-SVM+MVPO models. First, Fig. 3 illustrates the risk-return efficient frontiers for different models. In Fig. 3, the y -axis represents the portfolio expected return for the SVM+MVPO, LS-SVM+MVPO, ARIMA+MVPO, ARIMA+SVM+MVPO, and ARIMA+LS-SVM+MVPO models, while the x -axis presents the portfolio standard deviation. The graph distinctly shows that as the standard deviation value increases, the portfolio return value for different models also rises concurrently. Besides that, ARIMA+MVPO, SVM+MVPO, and LS-SVM+MVPO models become infeasible beyond a particular range of μ_{fix} values, whereas ARIMA+LS-SVM+MVPO and ARIMA+SVM+MVPO models offer workable solutions for all of the specified μ_{fix} values. Furthermore, the return of the ARIMA+LS-SVM+MVPO models demonstrates a more pronounced increase compared to the other models. Second, a comparison plot is illustrated in Fig. 4 by analyzing the optimal Sharpe ratio values of the different prediction models at different expected portfolio return values. It can be clearly noticed from the figure that the ARIMA+LS-SVM+MVPO model has higher Sharpe ratio values than the other models at most of the μ_{fix} values. Apart from that, the cumulative return graphs for different models at different μ_{fix} values are presented in Figs. 5–7 for better comparison between models. As can be seen, Figs. 5–7 represents the cumulative return graphs for different models at $\mu_{fix} = 0.0003, 0.0007, 0.0011$. In all the graphs, the x -axis represents the date, and the y -axis represents the cumulative returns of each model. It is clear from the graphs that there are significant increases in the cumulative return for all the models throughout all sub-graphs. However, the return values of the ARIMA+LS-SVM+MVPO model are higher than the other models in all the graphs.

5.4. Out of sample performance

This section offers a thorough examination of how well the suggested method performs by analyzing its behavior on out-of-sample data. This testing involves using an out-of-sample dataset, consisting of the same representative stocks listed in Table 1, gathered between July 1, 2023, and October 31, 2024. Consequently, portfolio returns are computed for the SVM+MVPO, LS-SVM+MVPO, ARIMA+MVPO, ARIMA+SVM+MVPO, and ARIMA+LS-SVM+MVPO models, with μ_{fix} values of 0.0007 and 0.0011. To provide clearer insights into the portfolio returns, Fig. 8 illustrates return graphs for different models

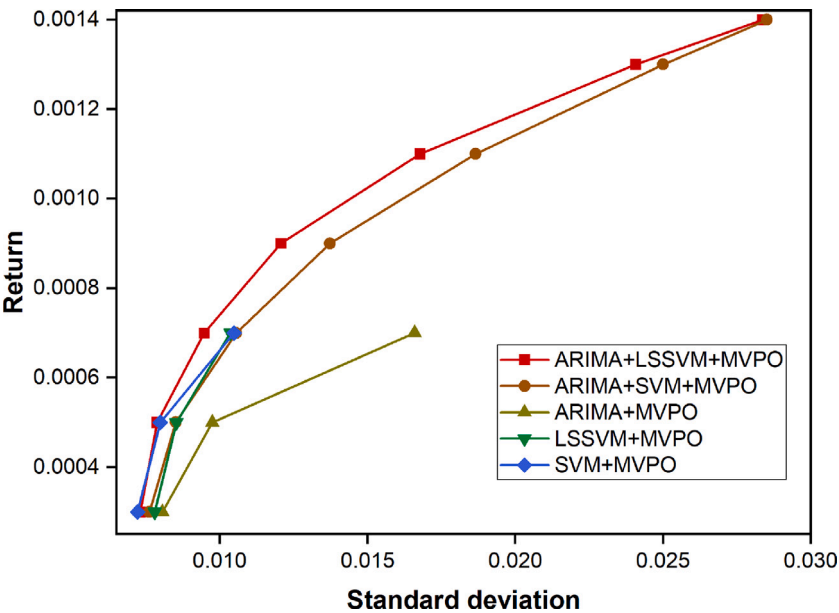


Fig. 3. Risk-return efficient frontier for different models.

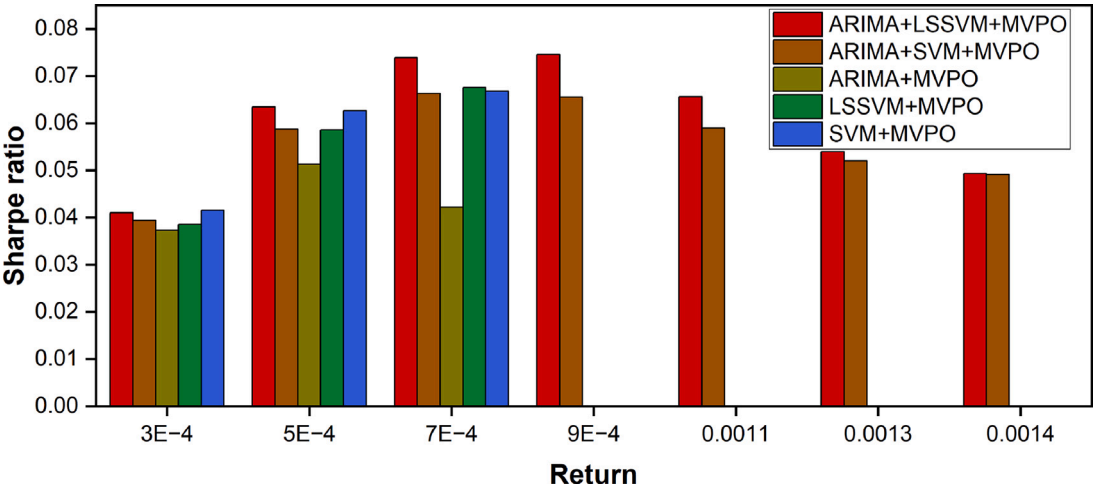


Fig. 4. Comparison of Sharpe ratio values of different models at different μ_{fix} values.

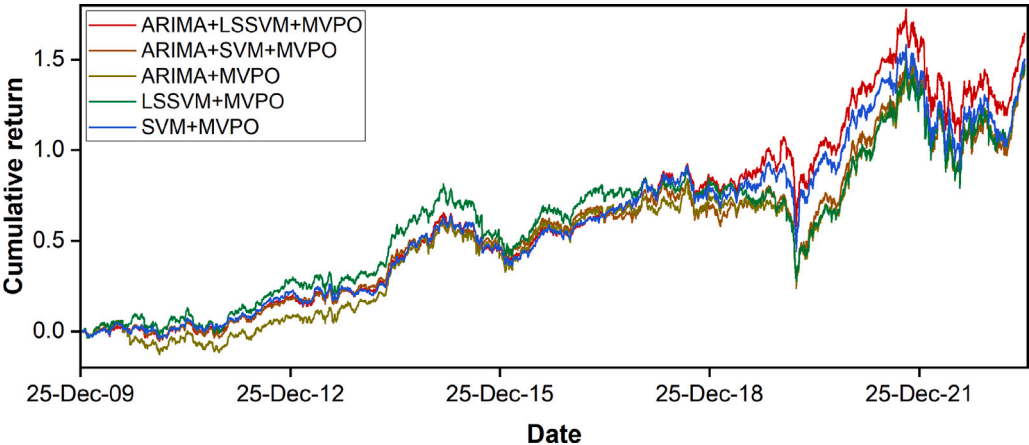
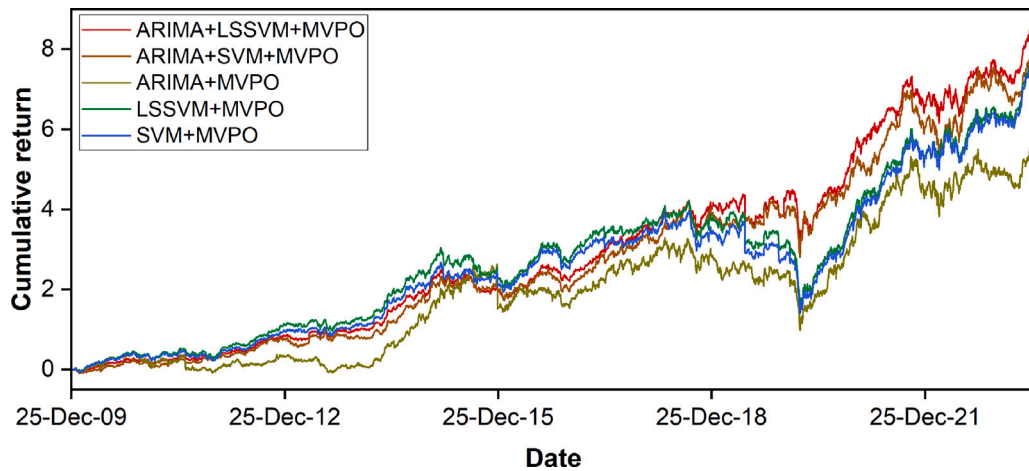


Fig. 5. Cumulative return for different model at $\mu_{fix} = 0.0003$.

Table 9

List of stocks selected from different prediction models.

SVM prediction	LS-SVM prediction	ARIMA prediction	ARIMA+SVM prediction	ARIMA+LS-SVM prediction
500 469	539 437	539 437	539 437	539 437
532 810	532 555	500 096	532 555	532 555
500 312	532 898	532 648	532 898	500 477
539 437	500 477	532 898	500 477	532 898
533 278	532 810	500 477	500 295	532 810
532 955	543 320	530 965	543 320	543 320
532 483	500 295	532 955	500 875	500 875
532 555	500 875	532 810	532 810	532 955
532 898	500 180	543 320	532 955	500 469
530 965	532 955	500 469	500 400	500 830
543 320	507 685	532 555	507 685	507 685
500 477	500 493	500 547	532 424	500 180
532 215	500 469	532 155	532 215	533 278
532 134	532 174	531 642	500 850	532 281
500 875	500 400	500 830	500 570	500 400
532 155	500 570	540 777	533 278	531 642
532 648	533 278	533 278	500 493	500 570
500 547	532 187	500 087	532 868	500 493
500 400	500 850	500 425	500 112	500 096
500 096	532 281	532 461	532 174	500 850
500 850	524 715	500 875	532 483	532 868
500 295	532 868	532 134	532 281	532 174
507 685	532 424	500 850	500 425	532 483
500 570	511 243	532 424	500 312	500 425
532 461	500 425	500 696	511 243	500 295
500 257	532 483	500 295	531 642	500 696
531 642	531 642	532 483	500 469	524 715
500 300	532 215	500 325	540 777	540 777
532 868	500 830	500 400	500 257	511 243
500 209	500 228	500 257	524 715	500 312

**Fig. 6.** Cumulative return for different model at $\mu_{fix} = 0.0007$.

under varying μ_{fix} values. The figure comprises two subgraphs, with the first illustrating portfolio return graphs for different models at $\mu_{fix} = 0.0007$, and the second subgraph representing portfolio return graphs at $\mu_{fix} = 0.0011$. Overall, analysis of these subgraphs indicates that the ARIMA+LS-SVM+MVPO model consistently outperforms the other models. Consequently, this research recommends employing the ARIMA+LS-SVM algorithm for constructing the MVPO model in daily trading investments.

5.5. Comparison results

In this section, comparison plots are presented to demonstrate the superiority of the proposed approach. Performance metrics, including standard deviation, Sharpe ratio, and cumulative return, are analyzed for the ARIMA+LS-SVM+MVPO model, Markowitz's equally weighted risk contribution (ERC) model by Maillard et al. [86], and the minimax portfolio optimization model by Young et al. [87]. Historical stock

prices from Table 1 are utilized, and the return, average return, and covariance of the assets are calculated. Markowitz's ERC model and the minimax portfolio optimization model are solved using LINGO software with the computed parameters. The performance metrics for various μ_{fix} values are compared with those of the ARIMA+LS-SVM+MVPO model, and the results are summarized in Table 10.

The table presents a comparison of the performance of three portfolio models—ARIMA+LS-SVM+MVPO, ERC Portfolio, and the minimax model by Young et al. across various expected return levels, evaluating standard deviation, variance, Sharpe ratio, and computational time. For every μ_{fix} level, the ARIMA+LS-SVM+MVPO model consistently achieves the lowest standard deviation and variance, indicating superior risk management compared to the ERC Portfolio and the model by Young et al. Additionally, higher Sharpe ratios in most cases demonstrate its ability to deliver better risk-adjusted returns. In terms of computational efficiency, both the ARIMA+LS-SVM+MVPO model and the model by Young et al. exhibit minimal computational

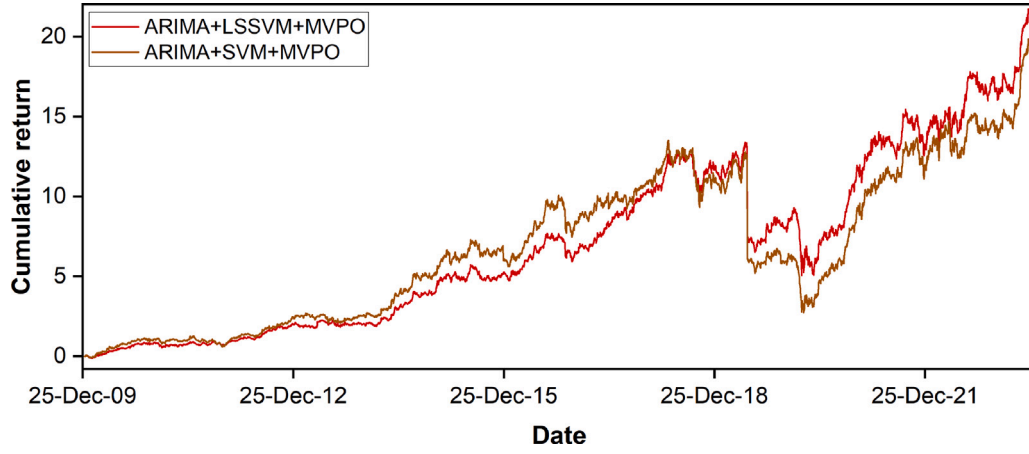
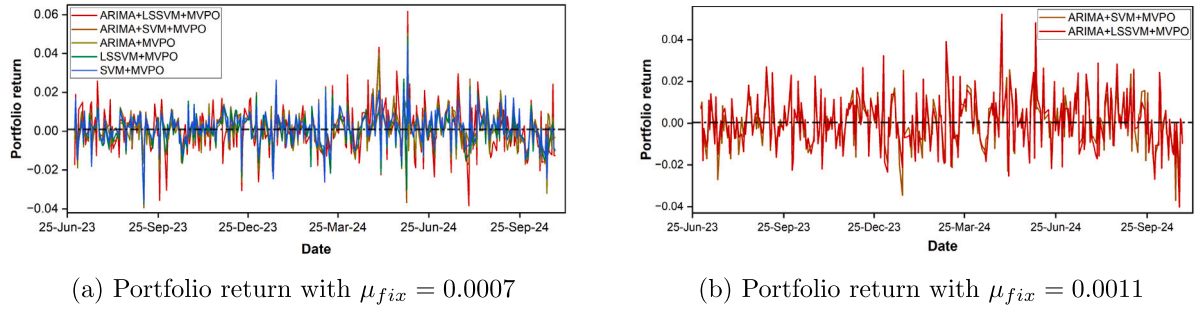
Fig. 7. Cumulative return for different model at $\mu_{fix} = 0.0011$.(a) Portfolio return with $\mu_{fix} = 0.0007$ (b) Portfolio return with $\mu_{fix} = 0.0011$ Fig. 8. Out of sample portfolio return with different μ_{fix} values.

Table 10

Comparison with existing literature.

μ_{fix}	Model	Standard deviation	Variance	Sharpe ratio	Time (s)
0.0003	Our approach	0.007313	0.000053	0.041023	0.00
	ERC portfolio	0.022468	0.000505	0.013352	0.34
	Young et al.	0.020304	0.000412	0.014775	0.00
0.0005	Our approach	0.007875	0.000062	0.063489	0.00
	ERC portfolio	0.028649	0.000821	0.017452	0.34
	Young et al.	0.028746	0.000826	0.017393	0.00
0.0007	Our approach	0.009472	0.000090	0.073904	0.00
	ERC portfolio	0.021923	0.000481	0.031930	0.33
	Young et al.	0.019650	0.000386	0.035624	0.00
0.0009	Our approach	0.012069	0.000146	0.074570	0.00
	ERC portfolio	0.022243	0.000495	0.040462	0.33
	Young et al.	0.017093	0.000292	0.052654	0.00
0.0011	Our approach	0.016770	0.000281	0.065595	0.00
	ERC portfolio	0.023184	0.000537	0.047447	0.33
	Young et al.	0.016979	0.000288	0.064785	0.01
0.0013	Our approach	0.024070	0.000579	0.054009	0.00
	ERC portfolio	0.024673	0.000609	0.052688	0.33
	Young et al.	0.023996	0.000576	0.054076	0.00

time, while the ERC Portfolio consistently requires 0.33–0.34 s, making it the slowest. These findings highlight the ARIMA+LS-SVM+MVPO model's effectiveness in balancing risk, return, and time, establishing it as the most robust option for portfolio optimization. Box plots in Figs. 9 and 10 further illustrate that the ARIMA+LS-SVM+MVPO model consistently demonstrates the lowest standard deviation and highest Sharpe ratio values across all μ_{fix} levels.

The memory usage of the three models in LINGO software is compared, measured in kilobytes (K). The ARIMA+LS-SVM+MVPO model

consumes 80K, indicating moderate memory usage. In contrast, the ERC Portfolio model is the most resource-intensive, requiring 786K, while the model by Young et al. demonstrates the highest memory efficiency, using only 44K. This comparison underscores the trade-off between resource efficiency and model complexity. The ARIMA+LS-SVM+MVPO model achieves a balance between memory efficiency and functionality, while the ERC Portfolio model, with its higher memory consumption, may cater to more complex tasks.

In addition, the cumulative return graphs for the models are analyzed at $\mu_{fix} = 0.0003$, 0.0007, and 0.0011 to evaluate portfolio return improvement. Figs. 11, 12, and 13 illustrate the cumulative returns at these levels, respectively. The results indicate that the ARIMA+LS-SVM+MVPO model consistently achieves the highest cumulative values, particularly under higher risk-return preferences. This suggests that the ARIMA+LS-SVM+MVPO model is well-suited for higher risk-return scenarios and is recommended for investors in the simulated trading environment.

6. Discussion and concluding remarks

This study demonstrates the effectiveness of a hybrid system combining time series forecasting and mean-variance portfolio management to build efficient portfolios. The proposed ARIMA+LS-SVM+MVPO model integrates ARIMA+LS-SVM for stock price prediction, evaluated using various metrics, with the MV model applied to top-performing stocks for investment strategies. Comparative analysis with other models and performance evaluation across multiple indicators validate its effectiveness. An experimental study using the Bombay Stock Exchange dataset highlights its advantages, summarized below.

This experiment evaluates the ARIMA+LS-SVM model's predictive performance against SVM, LS-SVM, ARIMA, and ARIMA+SVM using metrics such as MAE, MSE, and RMSE. The ARIMA+LS-SVM model

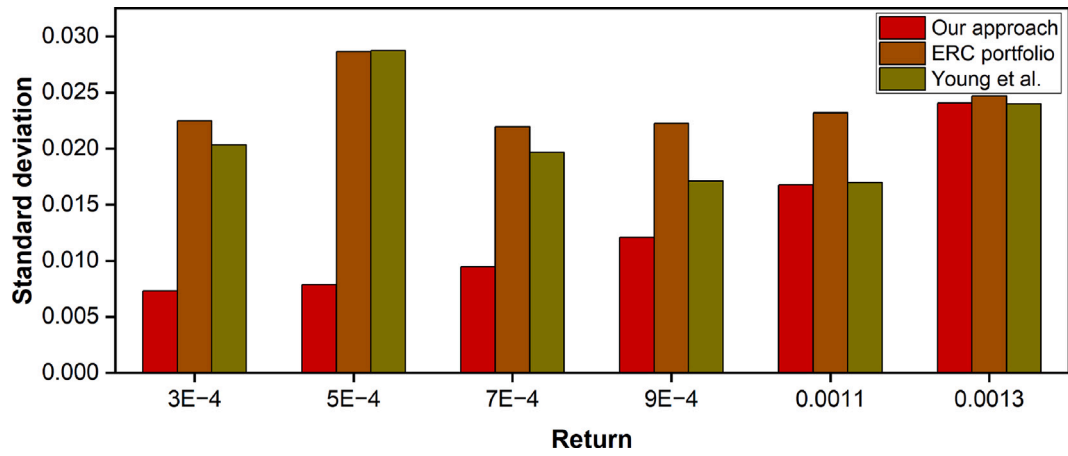


Fig. 9. Comparison of standard deviations with existing researches.

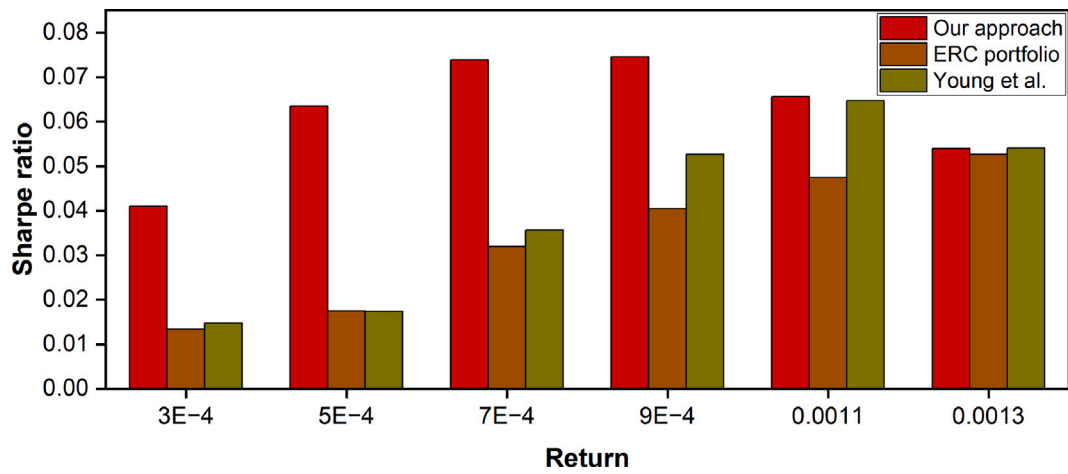


Fig. 10. Comparison of Sharpe ratios with existing researches.

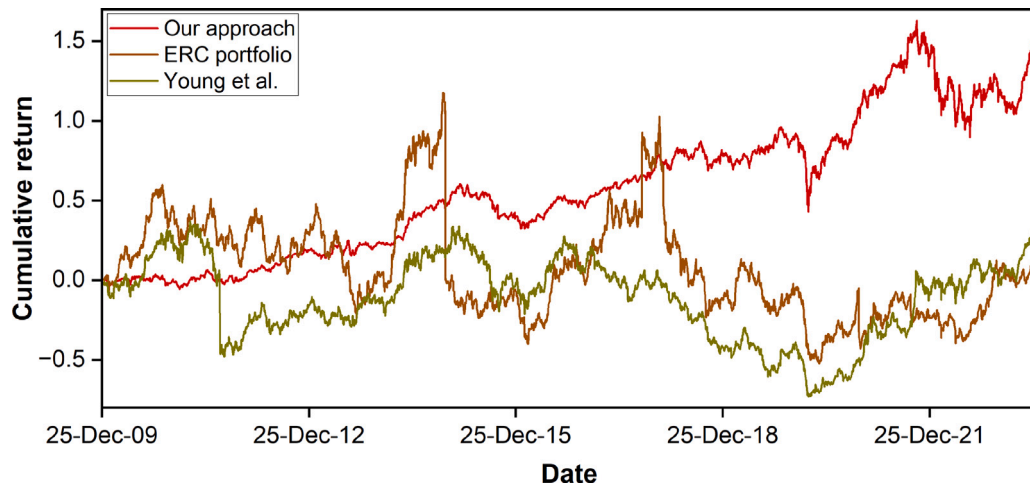


Fig. 11. Cumulative returns comparison for different model at $\mu_{fix} = 0.0003$.

demonstrates superior stock price prediction, achieving the lowest MSE and RMSE values among the models. Subsequently, the MV model is applied to optimize portfolios using stocks selected from the best-performing prediction model, resulting in efficient portfolios. The study further assesses the MV model combined with machine learning-based

predictions using performance metrics such as cumulative return, standard deviation, variance, Sharpe ratio, and average return. The results indicate that the ARIMA+LS-SVM+MVPO model achieves superior risk-return efficiency, delivering higher returns, lower variance, and standard deviation, a higher Sharpe ratio, and improved portfolio cumulative returns. Robustness is validated using an out-of-sample

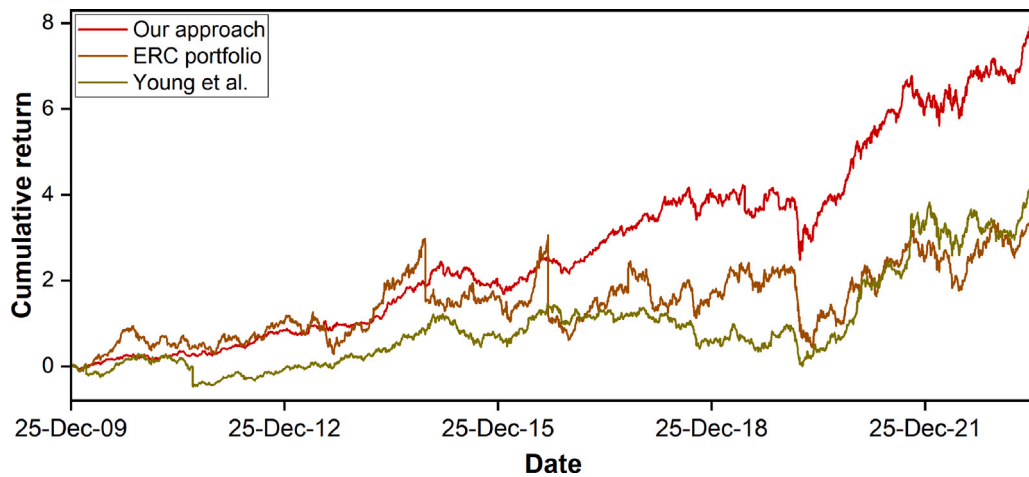


Fig. 12. Cumulative returns comparison for different model at $\mu_{fix} = 0.0007$.

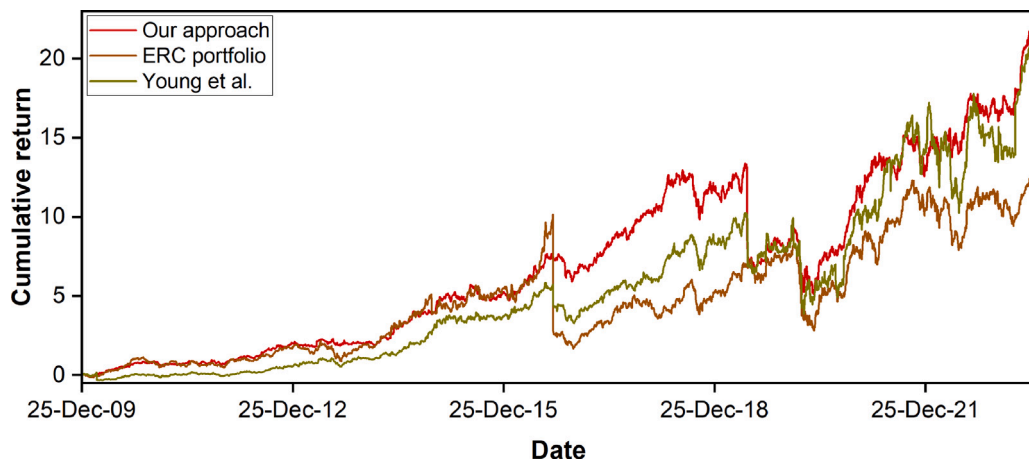


Fig. 13. Cumulative returns comparison for different model at $\mu_{fix} = 0.0011$.

dataset, where the ARIMA+LS-SVM+MVPO model consistently outperforms alternatives. Additionally, the proposed model is compared with two existing portfolio optimization methods, with results confirming its superiority across all performance indicators. These findings highlight the potential of the ARIMA+LS-SVM+MVPO model in applications such as algorithmic trading and risk management. While this study offers valuable insights, certain limitations should be acknowledged. Primarily, the reliance on historical closing prices as input features for stock price prediction may limit the model's comprehensiveness. Future research could explore the integration of additional factors such as news, investor sentiment, and economic conditions to enhance prediction accuracy. Furthermore, the utilization of only the BSE dataset restricts the scope of comparison; incorporating diverse stock market datasets could provide more robust results. Additionally, exploring various portfolio optimization models may enhance the out-of-sample performance of ARIMA+LS-SVM prediction-based portfolio optimization models.

CRedit authorship contribution statement

Jyotirmayee Behera: Writing – original draft, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Pankaj Kumar:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] H. Markowitz, Portfolio selection*, J. Finance 7 (1) (1952) 77–91.
- [2] P. Kumar, J. Behera, A. Bhurjee, Solving mean-VaR portfolio selection model with interval-typed random parameter using interval analysis, Opsearch 59 (1) (2022) 41–77.
- [3] Y. Song, G. Zhao, B. Zhang, H. Chen, W. Deng, W. Deng, An enhanced distributed differential evolution algorithm for portfolio optimization problems, Eng. Appl. Artif. Intell. 121 (2023) 106004.
- [4] Z. Hosseini-Nodeh, R. Khanjani-Shiraz, P.M. Pardalos, Portfolio optimization using robust mean absolute deviation model: Wasserstein metric approach, Finance Res. Lett. 54 (2023) 103735.
- [5] B. Li, Z. Lu, Uncertain random enhanced index tracking for portfolio selection with parameter estimation and hypothesis test, Chaos Solitons Fractals 168 (2023) 113125.
- [6] C. Sun, Q. Wu, X. Yan, Dynamic CVaR portfolio construction with attention-powered generative factor learning, J. Econom. Dynam. Control (2024) 104821.

- [7] Y. Ma, R. Mao, Q. Lin, P. Wu, E. Cambria, Quantitative stock portfolio optimization by multi-task learning risk and return, *Inf. Fusion* 104 (2024) 102165.
- [8] K. Su, Y. Yao, C. Zheng, W. Xie, Portfolio selection based on emd denoising with correlation coefficient test criterion, *Comput. Econ.* 63 (1) (2024) 391–421.
- [9] W. Wang, W. Li, N. Zhang, K. Liu, Portfolio formation with preselection using deep learning from long-term financial data, *Expert Syst. Appl.* 143 (2020) 113042.
- [10] C. Xiao, W. Xia, J. Jiang, Stock price forecast based on combined model of ARI-MA-LS-SVM, *Neural Comput. Appl.* 32 (2020) 5379–5388.
- [11] M. Zolfaghari, S. Gholami, A hybrid approach of adaptive wavelet transform, long short-term memory and ARIMA-GARCH family models for the stock index prediction, *Expert Syst. Appl.* 182 (2021) 115149.
- [12] S. Abolmakarem, F. Abdi, K. Khalili-Damghani, H. Didekhani, Predictive multi-period multi-objective portfolio optimization based on higher order moments: Deep learning approach, *Comput. Ind. Eng.* 183 (2023) 109450.
- [13] M. Ashrafzadeh, H.M. Taheri, M. Gharehgozlou, S.H. Zolfani, Clustering-based return prediction model for stock pre-selection in portfolio optimization using PSO-CNN+ MVF, *J. King Saud Univ.-Comput. Inf. Sci.* 35 (9) (2023) 101737.
- [14] M.K. Alamdari, A. Esfahanipour, H. Dastkhan, A portfolio trading system using a novel pixel graph network for stock selection and a mean-CDaR optimization for portfolio rebalancing, *Appl. Soft Comput.* 152 (2024) 111213.
- [15] B. Sahu, P. Kumar, Portfolio rebalancing model utilizing support vector machine for optimal asset allocation, *Arab. J. Sci. Eng.* (2024) 1–27.
- [16] J. Behera, P. Kumar, Implementation of machine learning in ℓ_∞ -based sparse sharpe ratio portfolio optimization: a case study on Indian stock market, *Oper. Res.* 24 (4) (2024) 62.
- [17] R. Thalheimer, M.M. Ali, Time series analysis and portfolio selection: An application to mutual savings banks, *South. Econ. J.* (1979) 821–837.
- [18] M. Aghagholizadeh Ardebili, M. Hashemi, A. Shahabi, M. Hatami Barough, Optimized selection of stock portfolio by using the fuzzy artificial neural networks web model, ARIMA & markowitz model in tehran stock exchange, *Eur. Online J. Natl. Soc. Sci.: Proc.* 4 (1 (s)) (2015) pp–831.
- [19] A.M. Rather, V. Sastry, A. Agarwal, Stock market prediction and portfolio selection models: a survey, *Opsearch* 54 (2017) 558–579.
- [20] T. Tarno, I.M. Di Asih, R. Rahmawati, A. Hoyyi, T. Trimono, M. Munawar, ARIMA-GARCH model and ARIMA-GARCH ensemble for value-at-risk prediction on stocks portfolio, 2020.
- [21] K. Raiysat, Portfolio Optimization Using ARIMA–Global Minimum Variance Approach (Ph.D. thesis), National College of Ireland, Dublin, 2020.
- [22] J.O. Fayomi, Application of Multi-Criteria Decision-Making Methods to Pension Portfolio Selection in Lithuania (Ph.D. thesis), Kauno technologijos universitetas, 2021.
- [23] J. Lin, Improved Markowitz portfolio investment model based on ARIMA model and BP neural network, in: 2022 IEEE 2nd International Conference on Data Science and Computer Application, ICDSCA, IEEE, 2022, pp. 504–507.
- [24] V. Mihova, V. Centeno, I. Georgiev, V. Pavlov, Comparative analysis of ARIMA and modified differential equation approaches in stock price prediction and portfolio formation, in: *International Conference on New Trends in the Applications of Differential Equations in Sciences*, Springer, 2022, pp. 341–350.
- [25] M. Khashei, Z. Hajirahimi, A comparative study of series arima/mlp hybrid models for stock price forecasting, *Comm. Statist. Simulation Comput.* 48 (9) (2019) 2625–2640.
- [26] L. Rubio, K. Alba, Forecasting selected colombian shares using a hybrid ARIMA-SVR model, *Mathematics* 10 (13) (2022) 2181.
- [27] Q. Qin, Z. Huang, Z. Zhou, Y. Chen, W. Zhao, Hodrick–Prescott filter-based hybrid ARIMA-SLFNs model with residual decomposition scheme for carbon price forecasting, *Appl. Soft Comput.* 119 (2022) 108560.
- [28] F. Pokou, J. Sadefo Kamdem, F. Benhmad, Hybridization of ARIMA with learning models for forecasting of stock market time series, *Comput. Econ.* (2023) 1–51.
- [29] D. Alaminos, M.B. Salas, A. Patal-Ureña, Hybrid ARMA-GARCH-neural networks for intraday strategy exploration in high-frequency trading, *Pattern Recognit.* 148 (2024) 110139.
- [30] R. Alam, R.I. Arnob, A.E. Alam, An ARIMA-LSTM correlation coefficient based hybrid model for portfolio management of Dhaka Stock Exchange, in: *Emerging Technologies in Computing: Third EAI International Conference, ICETIC 2020*, London, UK, August 19–20, 2020, *Proceedings 3*, Springer, 2020, pp. 214–226.
- [31] Y. Hong, An empirical analysis on the portfolio of transnational auto market based on ARIMA-LSTM, in: 2022 2nd International Conference on Financial Management and Economic Transition (FMET 2022), Atlantis Press, 2022, pp. 130–142.
- [32] J. Behera, A.K. Pasayat, H. Behera, COVID-19 vaccination effect on stock market and death rate in India, *Asia-Pac. Financ. Mark.* 29 (4) (2022) 651–673.
- [33] J.A. Suykens, J. Vandewalle, Least squares support vector machine classifiers, *Neural Process. Lett.* 9 (1999) 293–300.
- [34] M. Kadkhodazadeh, S. Farzin, A novel LSSVM model integrated with GBO algorithm to assessment of water quality parameters, *Water Resour. Manage.* 35 (2021) 3939–3968.
- [35] M. Cai, O. Hocine, A.S. Mohammed, X. Chen, M.N. Amar, M. Hasanipah, Integrating the LSSVM and RBFNN models with three optimization algorithms to predict the soil liquefaction potential, *Eng. Comput.* 38 (4) (2022) 3611–3623.
- [36] K. Zhang, K. Zhang, R. Bao, Prediction of gas explosion pressures: A machine learning algorithm based on KPCA and an optimized LSSVM, *J. Loss Prev. Process Ind.* 83 (2023) 105082.
- [37] M.A. Gandhi, V.K. Maharram, G. Raja, S. Sellapaandi, K. Rathor, K. Singh, A novel method for exploring the store sales forecasting using fuzzy pruning LS-SVM approach, in: 2023 2nd International Conference on Edge Computing and Applications, ICECAA, IEEE, 2023, pp. 537–543.
- [38] S. Maghsoudy, P. Zakerabbasi, A. Baghban, A. Esmaeili, S. Habibzadeh, Connectionist technique estimates of hydrogen storage capacity on metal hydrides using hybrid GAPSO-LSSVM approach, *Sci. Rep.* 14 (1) (2024) 1503.
- [39] Y. Li, Y. Yang, J. Wang, F. Liang, Thermal error modeling of servo axis based on optimized LSSVM with gray wolf optimizer algorithm, *Case Stud. Therm. Eng.* 53 (2024) 103858.
- [40] H. Sun, Y. Lu, A novel approach for solving linear fredholm integro-differential equations via LS-SVM algorithm, *Appl. Math. Comput.* 470 (2024) 128557.
- [41] S. Thanikodi, M. Meena, Y.D. Dwivedi, T. Aravind, J. Giri, M.S. Samdani, L. Kansal, M. Shahazad, M. Ilyas, Optimizing the selection of natural fibre reinforcement and polymer matrix for plastic composite using LS-SVM technique, *Chemosphere* 349 (2024) 140971.
- [42] D.-m. Xu, Z. Li, W.-c. Wang, An ensemble model for monthly runoff prediction using least squares support vector machine based on variational modal decomposition with dung beetle optimization algorithm and error correction strategy, *J. Hydrol.* 629 (2024) 130558.
- [43] L. Cui, Y. Xia, L. Lang, B. Hou, L. Wang, The dual mahalanobis-kernel LSSVM for semi-supervised classification in disease diagnosis, *Arab. J. Sci. Eng.* (2024) 1–19.
- [44] O. Hegazy, O.S. Soliman, M.A. Salam, Comparative study between FPA, BA, MCS, ABC, and PSO algorithms in training and optimizing of LS-SVM for stock market prediction, *Int. J. Adv. Comput. Res.* 5 (18) (2015) 35–45.
- [45] J. Chai, J. Du, K.K. Lai, Y.P. Lee, et al., A hybrid least square support vector machine model with parameters optimization for stock forecasting, *Math. Probl. Eng.* 2015 (2015).
- [46] I. Marković, M. Stojanović, J. Stanković, M. Stanković, Stock market trend prediction using AHP and weighted kernel LS-SVM, *Soft Comput.* 21 (2017) 5387–5398.
- [47] W. Zhang, Z. Wu, Optimal hybrid framework for carbon price forecasting using time series analysis and least squares support vector machine, *J. Forecast.* 41 (3) (2022) 615–632.
- [48] D. Zhao, J. Ding, S. Chai, Systemic financial risk prediction using least squares support vector machines, *Modern Phys. Lett. B* 32 (17) (2018) 1850183.
- [49] X.-L. Gong, X.-H. Liu, X. Xiong, X.-T. Zhuang, Forecasting stock volatility process using improved least square support vector machine approach, *Soft Comput.* 23 (2019) 11867–11881.
- [50] M. Kumar, M. Thenmozhi, Forecasting stock index returns using ARIMA-SVM, ARIMA-ANN, and ARIMA-random forest hybrid models, *Int. J. Bank. Account. Financ.* 5 (3) (2014) 284–308.
- [51] V. Vig, Time series forecasting using arima models: A systematic literature review of 2000s, *Comput. Intell. Appl. Softw. Eng. Probl.* (2023) 135–152.
- [52] Ü.Ç. Büyüksahin, Ş. Ertekin, Improving forecasting accuracy of time series data using a new ARIMA-ANN hybrid method and empirical mode decomposition, *Neurocomputing* 361 (2019) 151–163.
- [53] Y. Musa, S. Joshua, Analysis of ARIMA-artificial neural network hybrid model in forecasting of stock market returns, *Asian J. Probab. Stat.* 6 (2) (2020) 42–53.
- [54] Z. Hajirahimi, M. Khashei, A novel parallel hybrid model based on series hybrid models of ARIMA and ANN models, *Neural Process. Lett.* 54 (3) (2022) 2319–2337.
- [55] E. Alshawarbeh, A.T. Abdulrahman, E. Hussam, Statistical modeling of high frequency datasets using the ARIMA-ANN hybrid, *Mathematics* 11 (22) (2023) 4594.
- [56] Y. Shui-Ling, Z. Li, Stock price prediction based on ARIMA-RNN combined model, in: 4th International Conference on Social Science (ICSS 2017), 2017, pp. 1–6.
- [57] Z. Hajirahimi, M. Khashei, Weighted MLP-ARIMA series hybrid model for time series forecasting, *J. Ind. Eng. Manage. Stud.* 7 (2) (2020) 187–201.
- [58] V. Arangi, S.J.S. Krishna, T.N. Gongada, M. Basha, K. Santosh, S. Muthuperumal, Hybrid deep learning and time series analysis for stock market prediction: A multilayer perceptron and ARIMA modelling approach, in: 2024 10th International Conference on Advanced Computing and Communication Systems, ICACCS, 1, IEEE, 2024, pp. 421–427.
- [59] H.K. Choi, Stock price correlation coefficient prediction with ARIMA-LSTM hybrid model, 2018, arXiv preprint arXiv:1808.01560.
- [60] S. Kulshreshtha, et al., An ARIMA-LSTM hybrid model for stock market prediction using live data, *J. Eng. Sci. Technol. Rev.* 13 (4) (2020).
- [61] E. Dave, A. Leonardo, M. Jeanice, N. Hanafiah, Forecasting Indonesia exports using a hybrid model ARIMA-LSTM, *Procedia Comput. Sci.* 179 (2021) 480–487.
- [62] P. Lv, Q. Wu, J. Xu, Y. Shu, Stock index prediction based on time series decomposition and hybrid model, *Entropy* 24 (2) (2022) 146.
- [63] J. Zhang, H. Liu, W. Bai, X. Li, A hybrid approach of wavelet transform, ARIMA and LSTM model for the share price index futures forecasting, *N. Am. J. Econ. Financ.* 69 (2024) 102022.

- [64] C. Ma, J. Wu, H. Hu, Y.N. Chen, J.Y. Li, Predicting stock prices using hybrid LSTM and ARIMA model, *IAENG Int. J. Appl. Math.* 54 (3) (2024).
- [65] S. Saleti, L.Y. Panchumarthi, Y.R. Kallam, L. Parchuri, S. Jitte, Enhancing forecasting accuracy with a moving average-integrated hybrid ARIMA-LSTM model, *SN Comput. Sci.* 5 (6) (2024) 704.
- [66] Y. Si, S. Nadarajah, Z. Zhang, C. Xu, Modeling opening price spread of shanghai composite index based on ARIMA-GRU/LSTM hybrid model, *PLoS One* 19 (3) (2024) e0299164.
- [67] F.H. Mustapa, M.T. Ismail, Modelling and forecasting S&P 500 stock prices using hybrid Arima-Garch Model, in: *Journal of Physics: Conference Series*, Vol. 1366, IOP Publishing, 2019, 012130.
- [68] S. Setyowibowo, M. As'ad, S. Sujito, E. Farida, Forecasting of daily gold price using ARIMA-GARCH hybrid model, *J. Ekon. Pembang* 19 (2) (2022) 257–270.
- [69] N. Vo, R. Ślepaczuk, Applying hybrid ARIMA-SGARCH in algorithmic investment strategies on S&P500 index, *Entropy* 24 (2) (2022) 158.
- [70] A.I. Adewole, On the hybrid of Arima and Garch model in modelling volatilities in Nigeria stock exchange, *Bima J. Sci. Technol.* 8 (2A) (2024) 169–180.
- [71] D.S.d.O.S. Júnior, J.F. de Oliveira, P.S. de Mattos Neto, An intelligent hybridization of ARIMA with machine learning models for time series forecasting, *Knowl.-Based Syst.* 175 (2019) 72–86.
- [72] Y. Yusof, Z. Mustaffa, A review on optimization of least squares support vector machine for time series forecasting, *Int. J. Artif. Intell. Appl.* 7 (2) (2016) 35–49.
- [73] X. Yan, N.A. Chowdhury, Mid-term electricity market clearing price forecasting: A hybrid LSSVM and ARMAX approach, *Int. J. Electr. Power Energy Syst.* 53 (2013) 20–26.
- [74] W. Sun, J. Zhang, A novel carbon price prediction model based on optimized least square support vector machine combining characteristic-scale decomposition and phase space reconstruction, *Energy* 253 (2022) 124167.
- [75] X. Chen, X. Wei, Z. Su, et al., Portfolio investment algorithm based on LSTM-ARIMA prediction model, *Autom. Mach. Learn.* 3 (1) (2022) 22–32.
- [76] K. Kashif, R. Ślepaczuk, Lstm-arima as a hybrid approach in algorithmic investment strategies, 2024, arXiv preprint [arXiv:2406.18206](https://arxiv.org/abs/2406.18206).
- [77] B. Katarina, Gunardi, Optimization of Bayesian structural time series (BSTS) applications in forecasting stock prices through state components selection, in: *International Conference on Educational Technology and Administration*, Springer, 2022, pp. 229–248.
- [78] P. Wang, H. Zhang, Z. Qin, G. Zhang, A novel hybrid-garch model based on ARIMA and SVM for PM2. 5 concentrations forecasting, *Atmos. Pollut. Res.* 8 (5) (2017) 850–860.
- [79] C. Ordóñez, F.S. Lasheras, J. Roca-Pardiñas, F.J. de Cos Juez, A hybrid ARIMA–SVM model for the study of the remaining useful life of aircraft engines, *J. Comput. Appl. Math.* 346 (2019) 184–191.
- [80] J. Wu, Y.-G. Wang, H. Zhang, Augmented support vector regression with an autoregressive process via an iterative procedure, *Appl. Soft Comput.* 158 (2024) 111549.
- [81] I. Muchtadi-Alamsyah, R. Viltoriano, F. Harjono, M. Nazaretha, M. Susilo, A. Bayu, B. Josaphat, A. Hakim, K. Syuhada, Support vector regression-based heteroscedastic models for cryptocurrency risk forecasting, *Appl. Soft Comput.* (2024) 111792.
- [82] J.A. Suykens, J. De Brabanter, L. Lukas, J. Vandewalle, Weighted least squares support vector machines: robustness and sparse approximation, *Neurocomputing* 48 (1–4) (2002) 85–105.
- [83] S. Karthika, V. Margaret, K. Balaraman, Hybrid short term load forecasting using ARIMA-SVM, in: *2017 Innovations in Power and Advanced Computing Technologies (I-PACT)*, IEEE, 2017, pp. 1–7.
- [84] L. Yuan, Stock index forecasting based on hybrid ARIMA and LSSVM methodology, in: *2015 International Conference on Education, Management, Information and Medicine*, Atlantis Press, 2015, pp. 679–684.
- [85] J. Behera, A.K. Pasayat, H. Behera, P. Kumar, Prediction based mean-value-at-risk portfolio optimization using machine learning regression algorithms for multi-national stock markets, *Eng. Appl. Artif. Intell.* 120 (2023) 105843.
- [86] S. Maillard, T. Roncalli, J. Teiletche, The properties of equally weighted risk contribution portfolios, *J. Portf. Manag.* 36 (4) (2010) 60.
- [87] M.R. Young, A minimax portfolio selection rule with linear programming solution, *Manage. Sci.* 44 (5) (1998) 673–683.